EXCESSIVE SHIFT SHOCK FROM N TO D OR N TO R POSITION [CW6A-EL]

id050321102000

TROUBLESHOOTING ITEM	Excessive shift shock from N to D or N to R position
DESCRIPTION	• Strong shock is felt when shifting from N to D or N to R position at idle.
POSSIBLE CAUSE	1. Insufficient TCM learning (Insufficient learning in N to D position or N to R position) 2. Not within idle speed specification 3. Signal malfunction — APP sensor malfunction — CKP sensor malfunction 4. Poor installation or worn of engine mount, suspension

Diagnostic procedure

STEP	INSPECTION		ACTION
1	**VERIFY MALFUNCTION SYMPTOM** • Is the malfunction symptom (shift shock/shifting time) at the same level compared to a vehicle of the same model?	Yes	Symptom troubleshooting is completed.
		No	Go to the next step.
2	**PERFORM ON-BOARD DIAGNOSTIC TO VERIFY HYDRAULIC PRESSURE SWITCH OPERATION PRESSURE** • Is the operation pressure normal?	Yes	Go to the next step.
		No	Replace the control valve body. (See CONTROL VALVE BODY REMOVAL/INSTALLATION [CW6A-EL].)
3	**PERFORM INITIAL LEARNING, N TO D POSITION LEARNING, AND N TO R POSITION LEARNING** Perform the Initial Learning. (See INITIAL LEARNING [CW6A-EL].) Perform N to D position learning using the following procedure: (1) The ATF temperature is **30 °C {86 °F} or more**. (2) Shift the selector lever from P position to N position. (3) Shift the selector lever from N position to D position and wait for **5 s or more** while holding the lever in the D position. (4) Shift the selector lever from D position to N position and wait for **5 s or more** while holding the lever in the N position. (5) Perform the procedures (3) to (4) above ten times. Perform N to R position learning using the following procedure: (1) The ATF temperature is **60 °C {140 °F} or more**. (2) Shift the selector lever from P position to N position. (3) Shift the selector lever from N position to R position and wait for **5 s or more** while holding the lever in the R position. (4) Shift the selector lever from R position to N position and wait for **5 s or more** while holding the lever in the N position. (5) Perform the procedures (3) to (4) above ten times. • Does the symptom disappear?	Yes	Symptom troubleshooting is completed.
		No	Go to the next step.
4	**INSPECT ENGINE CONTROL SYSTEM FOR MALFUNCTION** • Perform the symptom troubleshooting “NO.9 FAST IDLE/RUNS ON”. (See NO.9 FAST IDLE/RUNS ON [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is there any malfunction?	Yes	Repair or replace any malfunctioning parts according to the inspection result.
		No	Go to the next step.

STEP	INSPECTION		ACTION
5	**INSPECT SIGNAL PARTS FOR MALFUNCTION** • Inspect the value at the following PCM PIDs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) — APP (APP sensor) — RPM (CKP sensor) • Is there any malfunction?	Yes	Repair or replace any malfunctioning parts according to the inspection result.
		No	Go to the next step.
6	**INSPECT ENGINE MOUNT/SUSPENSION FOR POOR TIGHTENING** • Is there any malfunction?	Yes	Repair or replace any malfunctioning parts according to the inspection result. (See ENGINE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Symptom troubleshooting is completed.